Functional Differentiation in Policy Implementation Texts: Evidence from China's State Council during the 14th Five-Year Plan

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I. Introduction

Modern states confront an increasingly wide range of policy challenges, including economic development, environmental protection, social welfare provision, and public security management. Governing such diverse policy domains requires administrative systems enabled to address complex and heterogeneous tasks. A central insight in theories of modern governance is that states organize policy functions through differentiation across specialized domains and administrative actors. Modern governance systems rely on functional division of labor to manage complex and coordinate policy objectives rather than undifferentiated administrative structures.

In the context of China, however, the organization of state capacity is often interpreted primarily through the perspectives of political centralization. Existing studies frequently emphasize hierarchical authority, centralized leadership, and top-down policy coordination in the Chinese governance system.

A growing body of scholarship has examined Chinese policy texts to understand the priorities and agenda-setting patterns of the state. These studies typically analyze the distribution of policy themes in official documents, such as government work reports or planning documents, in order to identify shifts in policy attention over time. However, this literature has largely focused on what policy areas the state emphasizes rather than how governance functions are structurally organized within policy discourse.

This study addresses this gap by examining the functional structure of executive-level policy discourse in China during the 14th Five-Year Plan period (2021–2025). Using a corpus of approximately 130,000 paragraphs extracted from policy documents issued by the State Council and its ministries, the study applies supervised machine learning to classify paragraph-level policy content into five governance domains derived from the 14th Five-Year Plan framework. By reconstructing the distribution of governance functions at the paragraph

level, the analysis provides a systematic measurement of functional differentiation in policy implementation texts.

The empirical analysis yields four main findings. First, executive-level policy discourse exhibits a stable functional structure across governance domains, suggesting that policy implementation operates within a differentiated functional space. Second, this differentiation is reflected in the organizational structure of the administrative system: many institutions display concentrated attention to particular governance domains, indicating identifiable functional roles within the governance apparatus. Third, certain policy topics appear across multiple governance domains, suggesting the presence of cross-domain "connector" areas that integrate multiple governance objectives. Fourth, governance domains display systematic differences in discursive intensity, indicating that different functional areas of governance are associated with distinct textual styles and levels of policy elaboration. Taken together, these findings indicate that policy discourse reflects not only functional specialization and cross-domain coordination within the governance system, but also domain-specific patterns of policy expression.

This study makes three contributions to the literature on governance and policy analysis. First, it provides a systematic measurement of functional differentiation in executive-level policy discourse, demonstrating that policy texts can reveal structured governance functions. Second, methodologically, the study introduces a paragraph-level classification approach that allows governance structures to be reconstructed from large-scale policy corpora. Third, substantively, the findings contribute to debates on Chinese state capacity by showing that centralized political authority can coexist with differentiated functional governance.

II. Theoretical Framework

2.1 Functional Differentiation in Governance Systems

A central insight in sociological theories of modern governance is that complex societies are organized through functional differentiation. In Luhmann's theory of social systems, modern institutions evolve by separating governance tasks into different functional domains such as the economy, law, politics, and science (Luhmann, 1995).

Within the state apparatus, functional differentiation typically performs in the division of policy responsibilities across specialized administrative domains. Modern governments regulate economic development, environmental protection and social welfare through different policy areas, each characterized by specific regulatory instruments, institutional actors, and policy logic. This differentiation allows governance systems to manage heterogeneous policy problems through specialized mechanisms.

Existing studies of state capacity often infer functional differentiation from formal institutional arrangements. However, institutional structures alone may not fully reveal how governance functions are operationalized in practice. Policy discourse provides an alternative observational window. Executive-level policy texts translate broad political goals into operational rules, administrative procedures, and implementation directives.

Thus, if governance systems are functionally differentiated, executive-level policy discourse should reflect this differentiation.

2.2 Organizational Specialization and Bureaucratic Division of Labor

Functional differentiation at the systemic level is typically accompanied by organizational specialization. Classical theories of bureaucracy emphasize that modern administrative systems rely on division of labor to manage complex policy tasks. Weber's

analysis of bureaucratic organization highlights the importance of specialization, in which administrative units are assigned specific functional responsibilities (Weber, 1978).

In contemporary governance systems, ministries often develop relatively stable policy mandates corresponding to particular governance fields. Economic agencies oversee industrial development and macroeconomic coordination; environmental authorities regulate ecological protection; social-sector institutions administer welfare programs and public services. Through this specialization, administrative systems distribute policy responsibilities across organizational units.

If functional differentiation is embedded in administrative organization, policy discourse should exhibit systematic variation across institutions. Agencies responsible for specific governance domains should produce texts that concentrate on their respective policy fields, while cross-cutting institutions may display more diversified functional profiles.

2.3 Cross-field Coupling

Although functional differentiation forms governance systems, modern policy problems frequently span multiple domains. Policy process theories emphasize that governance often occurs within overlapping policy subsystems composed of actors, institutions, and ideas operating across issue areas (Sabatier and Jenkins-Smith, 1993).

Complex policy challenges often require coordination across different functions. As a result, certain policy topics may not be monopolized by a single governance domain. Instead, it may serve as connectors linking multiple functional areas.

This phenomenon can be conceptualized as cross-field coupling within a differentiated governance system. And such coupling reflects the integrative mechanisms required to coordinate complex policy objectives. Therefore, certain policy domains may

display diversified functional distributions, reflecting their role in linking multiple governance functions.

2.4 Discursive Consequences of Functional Differentiation

Functional differentiation may influence not only the allocation of policy attention but also the discursive form of policy communication. Different governance domains involve distinct regulatory logics and administrative practices, which may shape how policies are articulated in textual form.

Studies of policy design emphasize that policy instruments vary systematically across policy areas (Howlett, 2011). Regulatory domains may rely on formal rules and standardized clauses, while social policy often requires detailed descriptions of target groups, service mechanisms, and implementation arrangements.

These differences suggest that functional domains may exhibit distinct discursive characteristics in policy texts. Domains dealing with complex social programs may produce longer and more explanatory textual structures, whereas regulatory domains may rely on more concise and directive language

III. Research Design

3.1 Data and Unit of Analysis

This study examines policy documents issued by the State Council of the People's Republic of China and its affiliated ministries during the 14th Five-Year Plan period (2021–2025). These documents represent the executive-level implementation layer of the state apparatus. Unlike planning blueprints or political speeches, implementation documents specified regulatory procedures, administrative responsibilities, coordination mechanisms, and operational guidelines.

The full corpus was collected through automated web scraping of the official State Council policy document database. After removing non-policy materials, the corpus was segmented into paragraph-level textual units. The final dataset contains approximately 130,000 paragraphs.

The paragraph is adopted as the unit of analysis for two reasons. First, document-level topic tags reflect administrative categorization rather than functional content. A document classified under “Agriculture” may contain paragraphs concerning environmental protection, agricultural product price regulation, or farmland protection. Second, governance tasks are often operationalized at the paragraph level, with specific rules, tools, or objectives.

Therefore, paragraph-level segmentation provides a finer-grained representation of governance functions.

3.2 Construction of Functional Space

To measure functional differentiation, the study constructs a governance field structure based on the 14th Five-Year Plan (2021–2025). The Plan serves as the overarching institutional framework guiding policy formulation during the period under examination. Rather than adopting its full chapter structure, five abstract governance fields are derived:

- Innovation and Modern Industrial System
- Macroeconomics and Resource Allocation
- Green Development and Ecological Civilization
- Social Welfare and Social Development
- Public Safety

These five fields correspond to development capacity, macro-regulatory coordination, sustainability constraints, social reproduction, and risk governance. The classification

framework is institutionally anchored in the planning structure, ensuring alignment with the official governance structure.

3.3 Supervised Classification

To assign governance fields to each paragraph, a supervised multi-class classification approach was used.

A manually annotated training sample was constructed through paragraph-level coding. Each paragraph was assigned to one of the five governance fields based on its primary functional objective.

3.3.1 Feature Engineering

Two categories of features were included:

A. Textual Features

TF-IDF representations of paragraph text were used to capture semantic structure.

B. Structural Features

Policy classification variables were incorporated, including:

- document type (doc_type)
- first-level topic category (topic_lvl1)
- second-level topic category (topic_lvl2)
- dominant topic label

These structural features capture administrative metadata that may signal functional orientation.

3.3.2 Model Comparison

Two models were compared:

- Multinomial logistic regression with L1 regularization (glmnet)
- Linear Support Vector Machine (SVM)

Model performance was evaluated using cross-validation. Two evaluation metrics were reported: macro-averaged F1 score and relaxed accuracy. The relaxed accuracy metric treats predictions within semantically adjacent governance domains as partially acceptable, reflecting the theoretical proximity between certain fields.

The linear Support Vector Machine outperformed the multinomial logistic regression model. Specifically, the SVM achieved a macro F1 score of 0.559 and a relaxed accuracy of 0.701, compared to 0.470 and 0.617 for the glmnet model. Given its superior cross-validated performance, the SVM classifier was selected for full-sample prediction. The trained classifier was then used to predict governance fields for the full corpus of approximately 130,000 paragraphs.

3.4 Measurement Strategy

Based on paragraph-level predictions, several structural indicators were constructed.

3.4.1 Functional Share

For each year and each institution, the share of paragraphs assigned to each governance field was calculated. This provides a distributional representation of functional attention.

3.4.2 Organizational Concentration

To measure specialization, concentration indices were computed for each institution:

- Herfindahl-Hirschman Index (HHI)
- Entropy measures

3.4.3 Re-embedding Index

To capture cross-field coupling at the topic level, the study defines:

$$re_embedding_index = 1 - top_share$$

Where top_share represents the proportion of paragraphs within a topic assigned to the dominant governance field.

A high re_embedding_index indicates that a topic is distributed across multiple governance fields, suggesting cross-functional coupling.

3.4.4 Discursive Intensity

To assess differences in expression style across governance fields, paragraph length shares were compared with paragraph count shares:

$$intensity_diff = weighted_length_share - paragraph_share$$

Positive values indicate relatively longer textual expression within a governance field, while negative values indicate shorter, more compressed expression.

IV. Findings

4.1 Existence of Functional Structure

The first empirical question concerns whether executive-level policy text exhibits a stable functional structure. If governance functions are differentiated, paragraph-level predictions should reveal distributional patterns across years and policy themes rather than random or document-type-driven.

4.1.1 Annual Stability of Functional Distribution

Figure 1 presents the annual distribution of predicted governance fields from 2021 to 2025. Across the five-year period, the relative shares of the five governance domains remain

remarkably stable. Innovation and macroeconomic regulation consistently occupy substantial proportions of paragraph-level attention, while public safety and green development maintain distinct but smaller shares.

The absence of dramatic year-to-year fluctuation suggests that executive-level policy discourse is not event-driven in its structural composition. Instead, it reflects a relatively institutionalized allocation of functional attention.

This stability supports the expectation that governance functions are organized within a structured field rather than randomly distributed across texts.

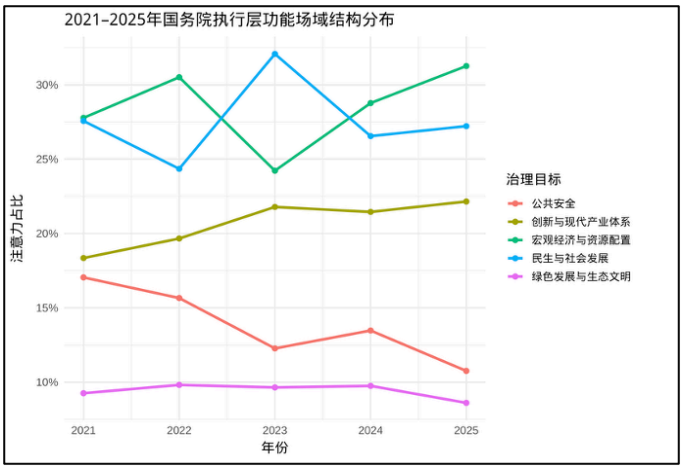

Figure 1: Distribution of Functional Field Structure of the State Council Executive Level, 2021-2025

4.1.2 Functional Differentiation Beyond Administrative Tags

To assess whether predicted governance fields merely replicate administrative classifications, the association between document-level topic categories and paragraph-level functional predictions was examined.

Cramér's V between document type and predicted domain is relatively low (0.19), indicating weak dependence. In contrast, the association between first-level policy topics (topic_lvl1) and predicted governance fields is substantially stronger (0.63).

This pattern suggests that paragraph-level functional differentiation is driven more by substantive governance tasks than by formal document categories. In other words, administrative labels do not determine functional allocation within texts.

4.2 Organizational Specialization

If functional differentiation is structurally embedded in executive-level discourse, it should not only perform as aggregate distributional stability but also as patterned variation across institutions. Specifically, different administrative units should exhibit identifiable functional concentrations reflecting organizational specialization.

4.2.1 Institutional Functional Profiles

Figure 2 presents the distribution of predicted governance fields for top 20 institutions. Each subplot represents the field attention structure of a single institution, measured as the share of paragraphs assigned to each governance domain.

A clear pattern emerges: many institutions correspond to a specific governance field. For example, ministries primarily responsible for industrial development exhibit strong concentration in the innovation and industrial system domain. Regulatory and macroeconomic ones show dominance in the macroeconomic and resource allocation field. Social-sector agencies demonstrate higher shares in social welfare and development.

This concentration suggests that paragraph-level predictions align with organizational responsibilities.

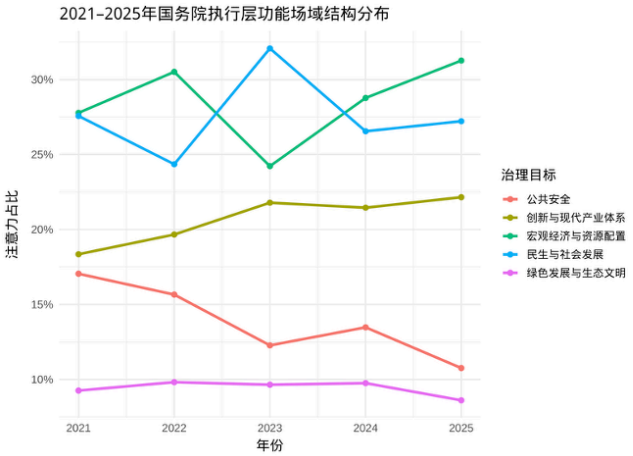

Figure 2: Institution-Field Attention Distribution

4.2.2 Concentration Measures

To quantify institutional specialization, concentration indices were calculated for each institution, including the Herfindahl-Hirschman Index (HHI) and entropy measures. Figure 3 presents the distribution of HHI scores across institutions with at least 300 paragraphs.

The red dashed line indicates the uniform-distribution benchmark (HHI = 0.2), corresponding to an equal allocation of attention across the five governance fields. As shown in the figure, the vast majority of institutions exhibit HHI values substantially above this benchmark. The distribution is strongly right-skewed, with many institutions clustering in the 0.6–1.0 range.

This pattern indicates that most institutions concentrate their policy discourse within a limited number of governance fields rather than distributing attention evenly across domains. In many cases, a single governance field dominates institutional output, producing single-peak or strong-dominant functional profiles.

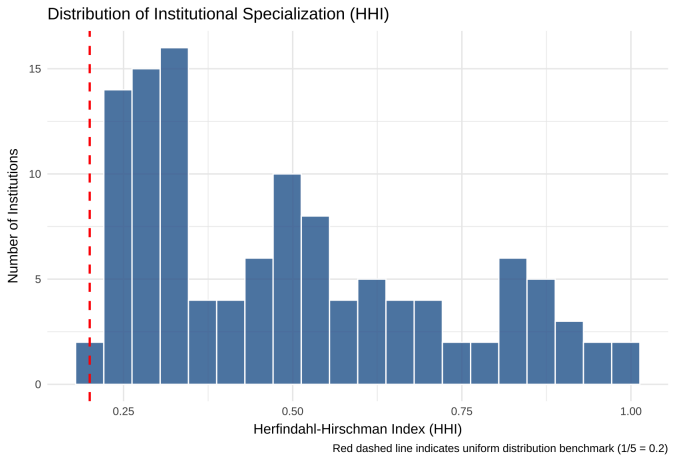

Figure 3: Distribution of Institutional Specialization (HHI)

4.3 Re-embedding and Cross-field Coupling

The evidence shows that executive governance exhibits clear patterns of functional differentiation. However, specialization at the organizational level does not totally explain the structural dynamics of policy attention. A relative question is: do policy domains remain isolated, or do they exhibit systematic cross-field embedding? To address this issue, the re-embedding index is examined.

4.3.1 Distribution of Re-embedding Across Themes

Figure 4 presents the distribution of the re-embedding index across all thematic categories, revealing substantial cross-theme variation in the degree of domain dispersion.

Several themes, including agriculture, forestry, and water resources; commerce, customs, and tourism exhibit comparatively high re-embedding scores. In these cases, attention is distributed across multiple governance domains rather than concentrated within a single functional field. That indicates that these themes are structurally embedded in diverse administrative logics.

By contrast, themes such as health and sports, national defense, and fiscal and financial administration display low re-embedding values. These policy areas are

characterized by strong concentration within a dominant governance domain, suggesting clear institutional ownership.

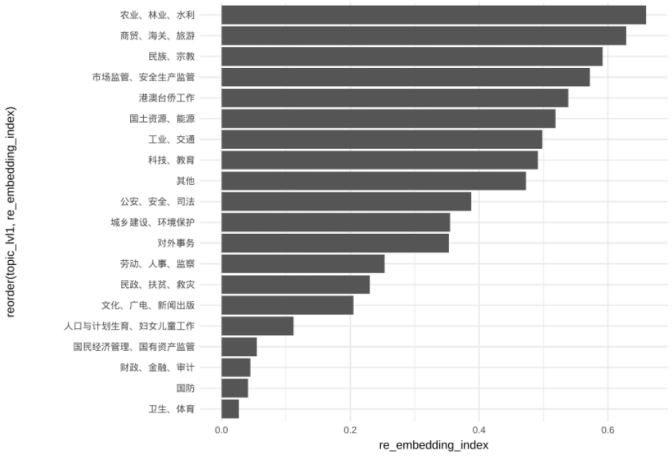

Figure 4: Distribution of re-embedding index across different themes

4.3.2 Structural Logic of Cross-field Dispersion

The variation observed across themes suggests that re-embedding is not random. Instead, it follows identifiable structural logics.

Themes with high re-embedding values typically share at least one of three characteristics. First, they involve functional interdependence, where policy implementation spreads economic, social, etc. Second, they require cross-sectoral coordination. Third, they may occupy politically salient positions.

Conversely, themes with low re-embedding values correspond to highly specialized domains with well-defined institutional mandates. In such cases, governance responsibility is concentrated, and cross-domain extension remains limited.

4.3.3 Differentiation and Structural Coupling

The coexistence of institutional specialization and theme-level dispersion reveals a layered governance architecture.

At the institutional level, executive agencies display clear domain concentration, indicating bureaucratic specialization. At the thematic level, however, certain policy areas circulate across multiple domains, generating horizontal linkages among specialized institutions.

This produces a system that is differentiated and interconnected. Organizational modules retain primary mandates, while attention allocation enables cross-domain coupling. Thus, Re-embedding allows the system to address complex and cross-cutting challenges without dissolving institutional boundaries.

4.4 Discursive Differentiation Across Governance Fields

Beyond structural stability, organizational specialization, and patterned re-embedding, executive-level policy texts exhibit systematic variation in discursive intensity across governance domains.

Figure 5 compares paragraph share and length-weighted share across the five governance fields from 2021 to 2025. The divergence between the two indicators captures differences in discursive intensity. When the length-weighted share exceeds paragraph share, a governance field is articulated through longer and more elaborated expressions. When paragraph share exceeds length share, the field is characterized by shorter, more compressed formulations.

Social Welfare and Social Development and Innovation and the Modern Industrial System display positive gaps between length share and paragraph share. In these domains, policies are expressed through longer and more detailed paragraphs. This pattern suggests that governance concerning social reproduction, welfare guarantees, and technological transformation requires extended articulation.

By contrast, Public Safety consistently exhibits a negative gap, with paragraph share exceeding length share. Policy expressions in this domain tend to be shorter, directive, and imperative in form.

Macroeconomics and Resource Allocation also demonstrates a tendency toward compressed expression. Although it occupies a substantial proportion of paragraphs, its length share remains comparatively lower. Macroeconomic governance frequently relies on target-setting, regulatory clauses, and principle-based formulations, which generate concise and standardized textual patterns.

Green Development and Ecological Civilization occupies an intermediate position. Length share slightly exceeds paragraph share in most years, indicating moderate elaboration. Environmental governance combines regulatory requirements with justificatory framing, producing neither highly compressed nor extensively elaborated discourse.

Taken together, these patterns indicate that functional differentiation is not limited to the distribution of attention across domains. It extends to the internal logic of policy articulation. Governance fields differ systematically in how attention is expressed: some domains are discursively expansive, others directive and condensed.

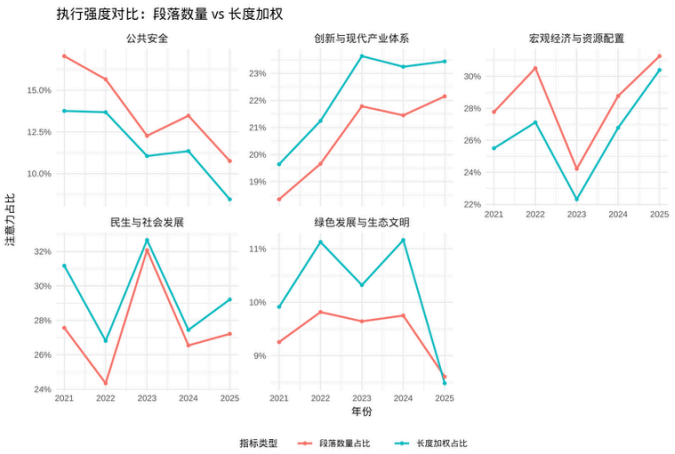

Figure 5: Text Intensity Difference

V. Discussion

The findings of this study contribute to ongoing debates about the role of policy discourse in revealing governance structures. Existing research on China's governance system often focuses on institutional arrangements or bureaucratic hierarchies. The evidence presented here suggests that governance structures can also be observed through patterns of policy discourse, providing a complementary empirical lens for understanding the organization of state capacity.

First, the stable distribution of governance domains in executive-level texts indicates that China's governance system exhibits functional differentiation despite its centralized political structure. Classical theories of bureaucracy suggest that hierarchical authority and functional specialization frequently coexist in modern administrative systems (Weber 1978). The evidence presented here suggests that centralized governance may operate through differentiated functional domains rather than through an undifferentiated administrative structure.

Second, the concentration of governance domains across institutions highlights the role of organizational specialization within the administrative system. Ministries appear to occupy relatively stable functional positions in the governance structure, and their policy texts reflect these institutional roles. In this sense, policy discourse provides observable traces of bureaucratic division of labor.

Finally, the presence of cross-domain policy topics suggests that functional differentiation does not imply rigid segmentation. Certain policy areas appear to connect multiple governance objectives, indicating the presence of integrative mechanisms within the governance system. Such cross-domain coupling is consistent with theories emphasizing coordination across policy subsystems when addressing complex governance challenges.

Taken together, the results suggest that policy discourse can serve as a useful empirical lens for examining how governance functions are structured within administrative systems. Rather than focusing solely on institutional design, future research may further explore how policy texts reflect the dynamic organization of governance tasks across policy domains and administrative actors.

Despite these contributions, several limitations should be acknowledged. First, the analysis relies on policy texts produced by the State Council and its ministries. While these documents represent the executive implementation layer, they may not fully capture the entire policy process or actual implementation outcomes. Second, the supervised classification approach inevitably introduces classification errors, even though cross-validation results indicate acceptable performance. Third, the functional field structure is derived from the 14th Five-Year Plan framework, which reflects an institutional perspective on governance domains. Alternative classifications may yield different field structures. Finally, the analysis focuses on the 2021–2025 period, and future research may examine longer time spans to explore how functional differentiation evolves across planning cycles.

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